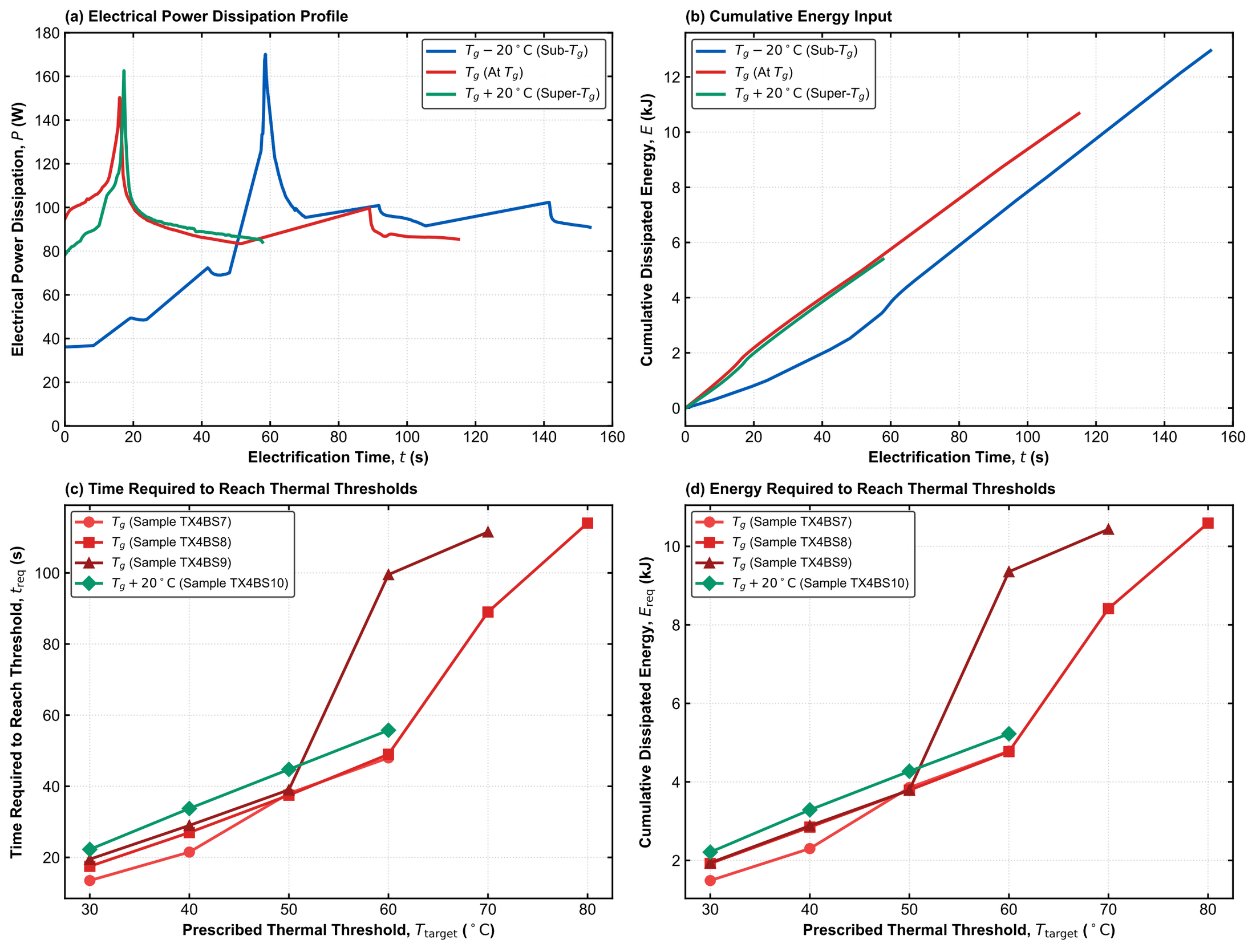
(a) Electrical Power Dissipation Profile
Electrical Power Dissipation, P (W)
Electrification Time, t (s)
T_g − 20 ˚C (Sub-T_g)
T_g (At T_g)
T_g + 20 ˚C (Super-T_g)
(b) Cumulative Energy Input
Cumulative Dissipated Energy, E (kJ)
Electrification Time, t (s)
T_g − 20 ˚C (Sub-T_g)
T_g (At T_g)
T_g + 20 ˚C (Super-T_g)
(c) Time Required to Reach Thermal Thresholds
Time Required to Reach Threshold, t_{req} (s)
Prescribed Thermal Threshold, T_{target} (˚C)
T_g (Sample TX4BS7)
T_g (Sample TX4BS8)
T_g (Sample TX4BS9)
T_g + 20 ˚C (Sample TX4BS10)
(d) Energy Required to Reach Thermal Thresholds
Cumulative Dissipated Energy, E_{req} (kJ)
Prescribed Thermal Threshold, T_{target} (˚C)
T_g (Sample TX4BS7)
T_g (Sample TX4BS8)
T_g (Sample TX4BS9)
T_g + 20 ˚C (Sample TX4BS10)